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Efficient and Lightweight Phishing Detection: A Case for Sustainable Cybersecurity with TF-IDF and LightGBM

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ABSTRACT Phishing is a damaging cyber threat in the digital landscape, and AI-generated attacks are becoming increasingly sophisticated. This paper presents a study on the development and evaluation of a computationally lightweight and efficient model for phishing email detection. An approach using TF-IDF + LightGBM was proposed, achieving 95.8% recall and 90.4% precision on a real-world dataset of 18,650 emails, reducing training time by 98% and memory requirements by 99% compared to state-of-the-art deep learning models (BERT, DistilBERT). Our methodology covers the complete machine learning pipeline, from data acquisition and exploratory analysis to rigorous hyperparameter tuning and performance evaluation. We demonstrate that an optimized classical ML approach can achieve performance comparable to deep learning models while being orders of magnitude more efficient. This work addresses the need for robust, efficient, and interpretable defenses that can be deployed in resource-constrained environments. We further contribute an analysis of model failures, feature importance, and robustness against adversarial attacks. Our findings demonstrate the value of democratizing access to advanced cybersecurity technologies through efficient and sustainable AI solutions.

INDEX TERMS Phishing, IA, Efficient, XAI, NLP, Multi-Model

I. INTRODUCTION

IN the digital ecosystem, phishing has been established as a significant attack vector, serving as a gateway to a broad spectrum of security incidents, ranging from credential theft and financial fraud to devastating ransomware attacks and corporate espionage [1]. The Anti-Phishing Working Group (APWG) reported over one million unique phishing attacks in the first quarter of 2025 alone, marking a persistent trend of increasing volume [2], [3], [4]. This deluge of malicious traffic is also rapidly becoming more sophisticated. The democratization of generative artificial intelligence (AI) and large language models (LLMs) has enabled adversaries to automate the creation of persuasive, contextually relevant, and grammatically flawless phishing lures at scale. According to recent security reports, AI-crafted phishing emails accounted for a substantial portion of campaigns in 2025, leveraging their realism to bypass traditional filters and deceive even discerning users [5], [6].

Unlike attacks that exploit purely technical vulnerabilities in software or network protocols, the enduring and evolving success of phishing lies in its fundamental exploitation of human psychology. The human element remains the most vulnerable component in the cybersecurity chain, with research consistently showing that cognitive and behavioral biases are the primary enablers of these attacks. Attackers masterfully leverage principles of social engineering, such as *authority bias* (impersonating a CEO or government official), *urgency* (creating a false sense of a need for immediate action), and *scarcity* (presenting a limited-time offer), to override an individual's rational judgment [7], [8]. A growing body of research demonstrates that factors such as cognitive load, personality traits, and an individual's perceived self-efficacy in identifying threats significantly influence susceptibility to deception [9], [10], [11]. This evidence shows that while user training and awareness are essential, they are insufficient and require additional robust technological safeguards.

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In response, the research community has developed a wide array of automated detection techniques. Early approaches relied on manually curated blacklists of malicious domains and keyword-based filters. While simple to implement, these methods are reactive and trivial for attackers to circumvent. The advent of machine learning (ML) marked a significant paradigm shift, enabling models to distinguish between legitimate and phishing emails using a variety of features. Traditional ML models, such as Support Vector Machines (SVM), Logistic Regression, and Random Forests, have demonstrated considerable success when trained on well-engineered features. More recently, the field has been dominated by deep learning (DL) models, particularly those based on the Transformer architecture, such as BERT (Bidirectional Encoder Representations from Transformers) [12] and its variants. These models have achieved state-of-the-art performance by capturing deep semantic and contextual relationships within the text, effectively understanding the nuances of language that characterize sophisticated phishing attacks [7], [13].

However, the pursuit of ever-higher accuracy has led to a significant challenge: the performance-efficiency dilemma [14], [15]. State-of-the-art DL models, while powerful, are resource-intensive. They require substantial computational power for training, often involving specialized hardware such as GPUs or TPUs, and their large model sizes and high inference latency make them challenging to deploy in real-time, resource-constrained environments, such as on-device email clients or local mail servers in small- to medium-sized enterprises (SMEs). This creates a gap: the most effective defenses are often inaccessible to those who need them most, leaving them reliant on older, less effective technologies. Furthermore, the black-box nature of these complex models raises significant concerns regarding interpretability and explainability (XAI), making their decision-making processes difficult to understand.

This paper directly addresses this by exploring the following research question: *Is it possible to develop a computationally lightweight and efficient phishing detection model that achieves performance comparable to state-of-the-art deep learning models, while maintaining a high degree of interpretability?*. We hypothesize that a carefully optimized pipeline combining a classic, efficient text representation with a modern, highly performant gradient-boosting framework can provide a practical and effective solution. Specifically, we propose and evaluate a model based on the Term Frequency-Inverse Document Frequency (TF-IDF) vectorization technique coupled with the Light Gradient Boosting Machine (LightGBM) classifier [16]. Our contributions are:

1) An end-to-end implementation and evaluation of a TF-IDF + LightGBM model for phishing detection, covering data preprocessing and feature engineering, hyperparameter tuning, and performance analysis.
2) A rigorous comparative analysis of our proposed model against both traditional ML baselines and a state-of-the-art Transformer-based model (DistilBERT), evaluating them on key metrics of accuracy, precision, recall, F1-score, training time, inference speed, and model size.
3) A demonstration that a lightweight, interpretable model can achieve performance that is highly competitive with complex deep learning models while being orders of magnitude more efficient in terms of computational resources, thereby presenting a viable and practical solution for real-world deployment.

This work reinforces the principle that, in the pursuit of advanced AI solutions, efficiency and practicality should not be sacrificed for marginal performance gains. de

The remainder of this paper is structured as follows: Section II provides a review of the relevant literature. Section III details the methodology, including the dataset, preprocessing, and models used. Section IV presents and analyzes the experimental results. Finally, Section V concludes the paper and outlines directions for future work.

II. LITERATURE REVIEW

Understanding the contemporary phishing detection landscape requires a rigorous examination of the coevolution of sophisticated attack vectors and their corresponding defenses [3]. This review is methodologically structured into three areas. First, we explore the technological evolution of detection methods, tracing their progression from rudimentary static, signature-based approaches to more dynamic heuristic-based systems and, ultimately, to advanced machine learning and deep learning models that can identify zero-day threats. Second, we analyze the state of the art, focusing on the trade-off between performance and efficiency. This involves evaluating how high-accuracy, computationally intensive models, such as deep neural networks, compare against more lightweight solutions in real-world, resource-constrained environments where low latency is essential. Third, we discuss feature engineering approaches, which remain fundamental to the efficacy of predictive models. This includes an examination of URL-based lexical features, webpage content analysis, and host-based attributes like DNS records. Collectively, these areas establish the foundational context for our research, which aims to propose novel solutions that address existing challenges in this dynamic domain.

A. THE EVOLUTION OF PHISHING DETECTION TECHNIQUES

The history of phishing detection is a classic cybersecurity arms race, with defenders continually developing new techniques to counter attackers' ever-evolving tactics. The progression can be broadly categorized into three main eras: manual/heuristic methods, classical machine learning, and deep learning.

Initially, defenses were rudimentary and reactive. Blacklisting was the most common approach, which involved maintaining lists of known malicious URLs, domains, and sender addresses. While simple and effective against known threats, blacklists are fundamentally flawed because they cannot generalize to unseen attacks. Attackers can easily circumvent them by registering new domains or using URL shortening

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services [3]. Complementing this was heuristic-based filtering, which relied on hand-crafted rules to identify suspicious content [1]. These rules would flag emails containing common phishing keywords, suspicious URL structures, or mismatches between link text and destination URLs. However, as attackers grew more sophisticated, they learned to craft messages that avoided these triggers, rendering heuristic methods less effective.

The introduction of classical machine learning (ML) marked a significant advancement. Instead of relying on rigid rules, ML models could learn to identify patterns of maliciousness from data. Researchers demonstrated the effectiveness of various algorithms, including Naive Bayes, Logistic Regression, Support Vector Machines (SVMs), and ensemble methods such as Random Forests [17]. These models were typically trained on a wide array of features, including lexical features from the email body, URL-based features, and header information. These methods improved detection rates and the ability to generalize to new attacks. However, they were heavily reliant on extensive, often complex feature engineering, a process that requires domain expertise and is time-consuming.

More recently, the field has been revolutionized by deep learning (DL), particularly advances in Natural Language Processing (NLP). Models based on Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were initially used to capture sequential information in text. However, the advent of the Transformer architecture [18] and models like BERT (Bidirectional Encoder Representations from Transformers) [19] established a new state of the art. Transformer-based models can understand language context and semantics to an unprecedented degree, enabling them to detect subtle, sophisticated phishing lures that mimic legitimate communication. Numerous studies have demonstrated the superior performance of models like BERT, RoBERTa, and their variants in phishing detection tasks [14], [17], [20]. These models eliminate the need for manual feature engineering by learning relevant features directly from raw text.

B. THE STATE OF THE ART AND THE EFFICIENCY GAP

While the performance of DL models is impressive, their success has come at a high cost, creating what we term the performance-efficiency dilemma. State-of-the-art Transformer models are massive, often containing hundreds of millions or even billions of parameters. Training these models is computationally intensive, requiring large datasets and specialized hardware such as GPUs or TPUs for extended periods. Their size and complexity also result in high inference latency, making them challenging to deploy in real-time applications or on resource-constrained devices such as mobile phones or edge servers [21].

This creates an accessibility gap. Large corporations with dedicated AI teams and extensive computational resources can leverage these powerful models, but small and medium-sized enterprises (SMEs) and individual users are often left behind. The high operational cost and complexity of deploying DL models make them impractical for many real-world scenarios. Furthermore, the "black-box" nature of these models raises significant concerns regarding explainability and interpretability (XAI). In a security context, it is often not sufficient to know that an email is malicious; it is also crucial to understand why it was flagged as such. This understanding is vital for forensic analysis, incident response, and building trust with end users. The opacity of complex DL models makes this level of interpretation difficult to achieve.

In response to this dilemma, a growing body of research has explored more efficient alternatives. Techniques such as model distillation and quantization have shown promise in reducing model size and improving speed [22]. Our work aligns with another vital research direction: exploring whether highly optimized classical ML algorithms can achieve reasonable performance for practical use while being more efficient. Gradient-boosting decision tree (GBDT) frameworks, in particular, have gained prominence. LightGBM [14], the algorithm used in this study, is a GBDT implementation known for its high training speed and low memory usage, making it an ideal candidate for creating lightweight yet powerful detection models.

C. FEATURE ENGINEERING AND TEXT REPRESENTATION

Regardless of the classification algorithm employed, the representation of email content is a component of any detection system. While some approaches incorporate a wide variety of metadata and structural features, these can sometimes be unreliable. For instance, header information can be easily forged by attackers. Therefore, the email body remains one of the most robust and reliable sources of information for detection.

For text representation, this study employs TF-IDF. TF-IDF is a classical NLP technique that has stood the test of time due to its simplicity, efficiency, and effectiveness. It creates a numerical representation of text by weighting words based on their frequency within a document and their rarity across the entire corpus. This allows the model to prioritize words that are highly indicative of a specific class. While TF-IDF does not capture the deep semantic relationships that Transformer embeddings do, its resulting feature vectors are sparse, high-dimensional, and highly interpretable. When paired with a robust, efficient classifier such as LightGBM, this classic approach can yield surprisingly strong results, as we demonstrate in this paper. With a focus on textual content, we aim to establish a strong baseline for a lightweight detector that is independent of potentially unreliable metadata.

III. METHODOLOGY

This section describes the methodology for automatic phishing detection in emails. Figure 1 presents the methods and procedures used to carry out phishing detection research in emails through natural language processing models. Advances in natural language models have enabled sophisticated threat-detection techniques for electronic communications, including phishing emails. However, using large models can

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be impractical in hardware-constrained or real-time environments.

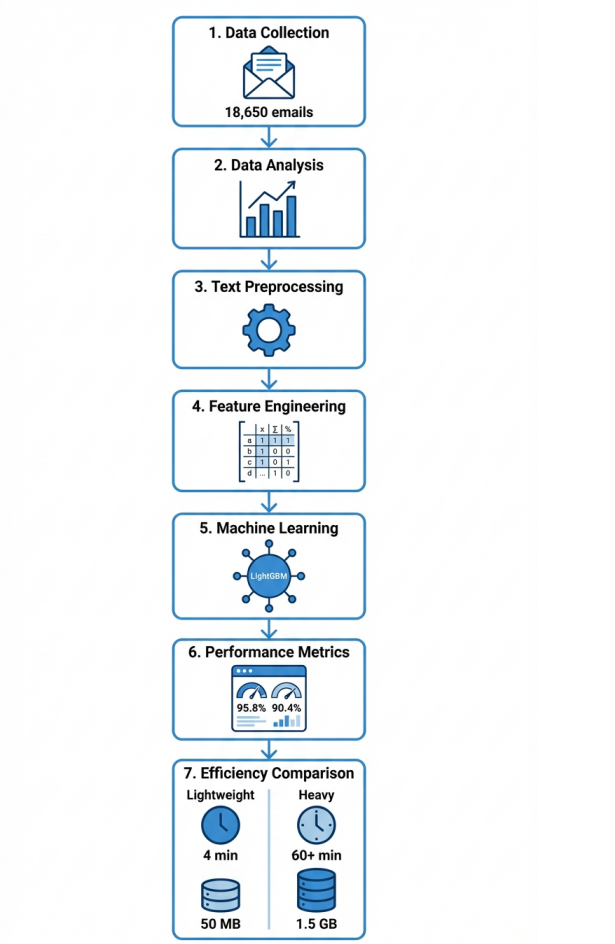


FIGURE 1. Methods and procedures used to carry out phishing detection research in emails through natural language processing models.

This section details each stage of our methodology, from data acquisition and characterization to model implementation and evaluation metrics.

A. DATA ACQUISITION AND CHARACTERIZATION

For this study, we constructed a corpus by aggregating data from multiple public sources to ensure a diverse collection of emails. The primary components are the Enron-Spam collection, which provides a large body of legitimate business and personal communications, and the Nazario phishing corpus, which contributes a set of real-world phishing examples. This aggregation strategy is relevant for capturing the heterogeneity of modern email traffic, from benign daily correspondence to sophisticated, evolving phishing tactics. The resulting dataset, comprising 18,650 emails, is statistically characterized in Table 1. The vocabulary size (125,432 unique tokens) and average email length emphasize the lexical complexity and information density of the data, necessitating advanced NLP models that capture deep semantic features rather than relying on superficial heuristics.

TABLE 1. Statistical Overview of the Aggregated Email Dataset

Metric	Value
Total Number of Emails	18,650
Phishing Emails	7,184 (38.5%)
Safe (Ham) Emails	11,466 (61.5%)
Total Vocabulary Size (Unique Words)	125,432
Average Words per Email	285
Average Characters per Email	1,624

For model training and evaluation, the dataset was partitioned via stratified random sampling into a training set (80%, or 14,920 emails) and a held-out test set (20%, or 3,730 emails). This stratification is relevant because it preserves the class distribution across both subsets, thereby mitigating distributional bias and ensuring that the test set serves as an unbiased proxy for evaluating the model's generalization on unseen data. The dataset exhibits moderate class imbalance: 61.5% of instances are legitimate, and 38.5% are phishing attempts. While not extreme, this level of imbalance necessitates careful consideration during evaluation, favoring metrics like the Precision-Recall curve over standard accuracy to avoid overly optimistic performance estimates skewed by the majority class, as shown in Figure 2.

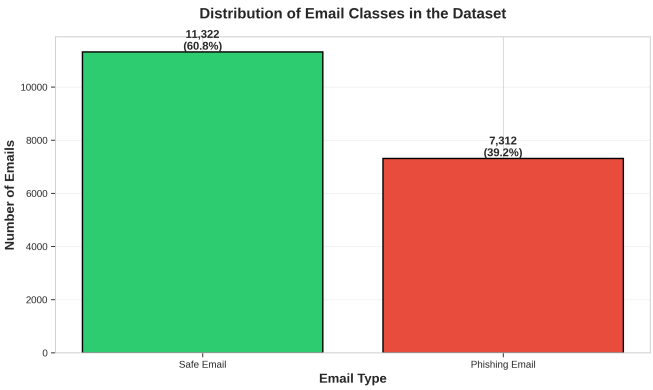


FIGURE 2. Distribution of email classes in the dataset.

A preliminary Exploratory Data Analysis (EDA) was conducted to elucidate the dataset's structural characteristics and identify salient discriminative features between the two primary classes of interest: legitimate emails and phishing emails. This initial analytical phase informs subsequent stages of feature engineering, feature selection, and machine learning model development, and reveals that the two email categories exhibit substantially divergent statistical properties and distributions. One of the most notable empirical observations emerged from the quantitative analysis of verbosity and content-length metrics. Figure 3 illustrates the probability distributions of character counts and word frequencies across both email classes, which represent fundamental, widely recognized metrics in Natural Language Processing (NLP) and computational linguistics applications.

The textual length of an email message can, in isolation, constitute a highly informative and discriminative feature; for

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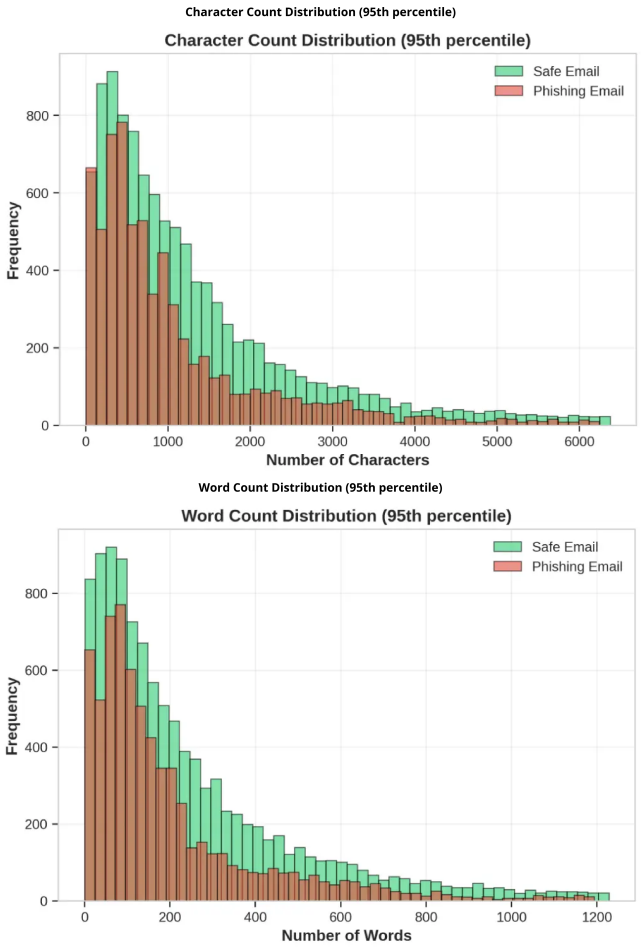


FIGURE 3. **Character and word count distributions for safe and phishing emails (95th percentile).**

instance, phishing emails may be characteristically brief to engender a sense of artificial urgency and temporal pressure on recipients, or conversely, may be substantially longer when incorporating elaborate and deceptive narratives designed to manipulate victims into compromising their security credentials.

To ensure optimal visualization clarity and the statistical representativeness of the distributional trends, the extreme tails of the empirical distributions were systematically truncated at the 95th percentile. This approach is fundamental for mitigating the effects of extreme outliers and anomalous data points, which would otherwise obscure subtle differences in the probability density functions. Furthermore, this truncation strategy facilitates the identification of modal patterns and central tendencies that better reflect typical email characteristics within each class. The distributional analysis reveals heterogeneous patterns across the two classes, suggesting that textual length metrics have considerable predictive utility for binary email classification. Additionally, the statistical dispersion and skewness of these distributions provide knowledge into the compositional strategies used by legitimate senders versus malicious actors, thereby establishing a robust empirical foundation for subsequent machine-learning feature extraction and model development.

The distributions indicate that phishing emails are more concise than legitimate emails. This is further confirmed in Figure 4, which shows the average counts. On average, legitimate emails are more than twice as long as phishing emails in terms of both character count (3,493 vs. 1,613) and word count (685 vs. 301). This suggests that attackers often favor brevity and directness to create a sense of urgency and prompt immediate action, whereas legitimate correspondence is frequently more detailed and context-rich.

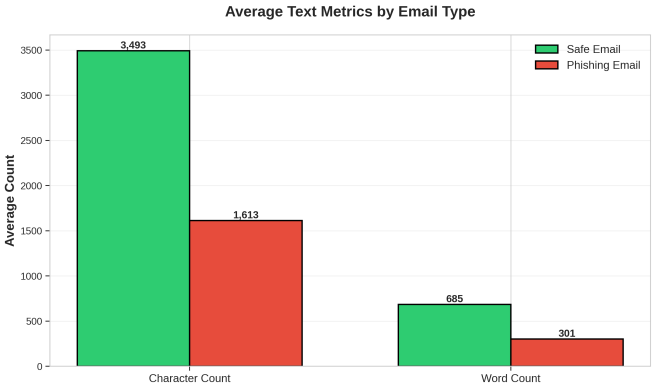


FIGURE 4. **Average character and word counts by email type.**

B. TEXT PREPROCESSING

Raw email text is unstructured and contains significant noise, such as HTML tags, special characters, and irrelevant headers, which can degrade model performance. Therefore, a preprocessing pipeline was implemented to clean and normalize the text. This pipeline consists of the following steps, applied sequentially to each email:

1) **Lowercasing:** All text was converted to lowercase. This is a standard normalization step that ensures the model treats words with different capitalization as a single token, reducing vocabulary size and improving generalization.
2) **Removal of memorable characters and punctuation:** All non-alphabetic characters, including numbers and punctuation, were removed. While punctuation can occasionally carry semantic weight, it often adds noise to bag-of-words models like TF-IDF, unnecessarily inflating the feature space.
3) **Tokenization:** The cleaned text was split into individual words or tokens. This process breaks the continuous text stream into discrete units that serve as the basis for our feature vectors.
4) **Stopword Removal:** Common English words that provide little discriminatory power were removed using the standard stopword list from the Natural Language Toolkit (NLTK) library. Removing these words helps the model focus on the more meaningful terms that distinguish phishing from legitimate content.

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5) **Lemmatization:** Words were reduced to their base or dictionary form (lemma). For example, "running," "ran," and "runs" were all converted to "run." This step consolidates the vocabulary by grouping related word forms into a single feature, thereby improving the model's ability to learn general patterns.

This preprocessing pipeline ensures that the subsequent feature engineering stage operates on clean, standardized, and meaningful textual data, which is essential for building an effective classifier.

C. MODEL TRAINING AND HYPERPARAMETER TUNING

To identify the most effective classifier for this task, four different machine learning models were trained and evaluated:

- **Logistic Regression:** A simple, fast, and highly interpretable linear model that serves as a strong baseline.
- **Random Forest:** An ensemble of decision trees that is robust to overfitting and generally provides high accuracy.
- **Support Vector Machine (SVM):** A robust model that finds an optimal hyperplane to separate the classes, using a linear kernel for efficiency.
- **LightGBM:** A high-performance gradient boosting framework optimized for speed and low memory usage.

These parameters, listed in Table 2, were selected based on best practices from the literature and empirical testing.

TABLE 2. Selected Hyperparameters for the LightGBM Classifier

Parameter	Value	Description
`objective`	`binary`	Binary classification
`metric`	`auc`	ROC AUC metric
`n_estimators`	1000	Max boosting rounds
`learning_rate`	0.05	Learning step size
`num_leaves`	31	Max leaves per tree
`max_depth`	-1	No depth limit
`min_child_samples`	20	Min samples in leaf
`feature_fraction`	0.9	Feature subsampling
`bagging_fraction`	0.8	Data subsampling
`bagging_freq`	5	Bagging frequency
`verbose`	-1	Suppress output
`n_jobs`	-1	Use all CPU cores

For the LightGBM model, the focus of this study, a set of optimized hyperparameters was used to balance performance and efficiency. Each model was trained on the training set and evaluated on the held-out test set. The primary selection metric was the recall of the positive class. The choice of recall is critical in fraud and security applications, as it prioritizes minimizing false negatives, which pose the most significant potential harm to the end user. Other metrics, including accuracy, precision, and F1-score, were also calculated for an evaluation.

D. FEATURE ENGINEERING: TF-IDF

Feature engineering is the process of transforming raw data into numerical representations that are understandable to machine learning algorithms. For this text-based task, we used TF-IDF vectorization. Unlike opaque embeddings from deep learning models, TF-IDF features have a direct correspondence to words, making the model's decisions easier to interpret.

TF-IDF assigns a weight to each word in a document, reflecting its importance. The weight is the product of two metrics:

- Term Frequency (TF): This measures how frequently a term appears in a document. It is often normalized to prevent a bias towards longer documents.
- Inverse Document Frequency (IDF): This measures how important a term is across the entire corpus. It penalizes common words and gives higher weight to rare words across documents, making them more discriminative.

The final TF-IDF score for a term t in a document d from a corpus D is given by:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D) \tag{1}$$

For our implementation, we used the 'TfidfVectorizer' from the Scikit-learn library with the following key parameters:

For our implementation, we used the `TfidfVectorizer` from the Scikit-learn library with the following key parameters:

- `max_features = 5000`: We limited the vocabulary to the top 5,000 most frequent terms across the corpus. This strikes a balance between capturing sufficient information to discriminate between classes and maintaining computational efficiency by controlling the dimensionality of the feature space.
- `ngram_range = (1, 2)`: We included both unigrams and bigrams as features. Bigrams are relevant for capturing local context and phrases that are more informative than individual words.

Following this process, each email in the dataset is transformed into a 5,000-dimensional sparse vector, where each dimension corresponds to the TF-IDF score of a specific unigram or bigram. This representation is then fed into our classification model.

E. CLASSIFICATION MODEL

For the classification task, we selected the Light Gradient Boosting Machine (LightGBM), a high-performance gradient boosting framework. LightGBM was chosen over other popular boosting algorithms, such as XGBoost, primarily for its superior efficiency. This results in significantly faster training speeds and lower memory consumption, making it an ideal choice for our goal of developing a lightweight model.

We compared the performance of LightGBM against three other widely used machine learning models to serve as baselines: Logistic Regression, a simple linear model; Random Forest, an ensemble model known for its robustness; and a linear Support Vector Machine (SVM), a classifier effective in high-dimensional spaces. Across all models, we performed

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hyperparameter tuning with a 5-fold cross-validation grid search to identify the optimal parameter configuration.

To provide a balanced assessment of model performance, we used four standard classification metrics:

- Accuracy: The proportion of correctly classified emails. While intuitive, it can be misleading on imbalanced datasets.
- Precision: The proportion of predicted phishing emails that were actually phishing. High precision is essential to minimize false positives, which can lead to legitimate emails being incorrectly blocked.
- Recall (sensitivity): The proportion of actual phishing emails that were correctly identified. High recall is critical for security applications, as it reflects the model's ability to catch as many threats as possible.
- F1-Score: The harmonic mean of precision and recall. It provides a single score that balances precision and recall.

In phishing detection, there is a trade-off between precision and recall. A false negative can have severe consequences, such as financial loss or a system breach. A false positive is an inconvenience but is generally less damaging. Therefore, while the F1-score provides a balanced view, recall is the more critical metric for this specific application.

IV. RESULTS AND DISCUSSION

Evaluation of the proposed TF-IDF with the LightGBM model resulted in efficient performance metrics that align with the objectives of creating a lightweight, low-cost, and accurate detection solution. This section presents the quantitative results, discusses their implications qualitatively, and contextualizes the findings within the broader cybersecurity landscape.

A. QUANTITATIVE MODEL PERFORMANCE

Performance metrics are essential for validating the effectiveness of any classifier, particularly in high-risk domains such as cybersecurity, where the cost of errors can be substantial. The models were evaluated on a held-out test set comprising 20% of the original dataset (3,730 emails). Table 3 provides a detailed comparison of the performance of the four evaluated machine learning models.

TABLE 3. Comparative Performance of Machine Learning Models on the Test Set

Model	Acc.	Prec.	Rec.	F1
Log. Regression	92.7%	89.2%	92.5%	90.8%
Random Forest	94.4%	90.8%	95.3%	93.0%
LightGBM	**94.4%**	**90.4%**	**95.8%**	**93.1%**
SVM (Linear)	92.4%	88.0%	93.4%	90.6%

The performance of the four candidate models, namely Logistic Regression, Random Forest, LightGBM, and Support Vector Machine (SVM), was rigorously evaluated on the held-out test set. As illustrated in Figure 5 and detailed in Table 3, the LightGBM model achieved superior performance, attributable to its gradient-based optimization and efficient leaf-wise tree growth. Although Random Forest also demonstrated competitive results, as expected from a robust ensemble method, LightGBM consistently outperformed it. One noteworthy outcome is the recall of 95.8% achieved for the phishing class. This metric is essential because it indicates that the model successfully identified nearly 96 out of every 100 phishing instances, thereby minimizing false negatives. Such high sensitivity is critical to the primary objective of preventing malicious emails from reaching end users, thereby directly reducing residual risk and the potential attack surface. Furthermore, the model achieved robust precision of 90.4%, resulting in minimal false positives. This balance is essential for maintaining user trust and operational continuity, preventing unnecessary workflow disruptions and alert fatigue. The resulting high F1-score further substantiates the model's robustness, indicating a well-calibrated equilibrium between precision and recall. In contrast, the linear nature of Logistic Regression and the kernel-based nature of SVMs likely limited their ability to model intricate nonlinear patterns in feature space. Figure 5 provides a comparative visualization, confirming LightGBM's superior performance across all key evaluation metrics.

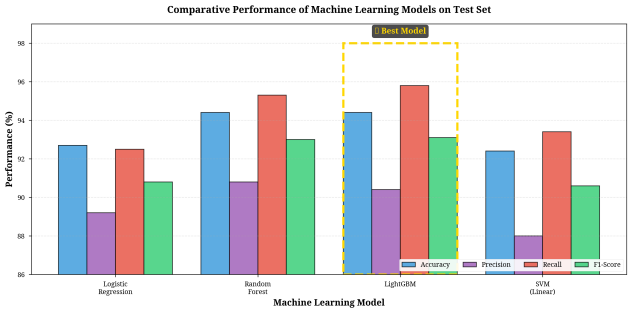


FIGURE 5. Comparative Performance of the Four Candidate Models. LightGBM (green) achieves the highest scores across all metrics, particularly excelling in Recall (95.8%) and F1-Score (93.1%).

Given the class imbalance in the dataset, conventional metrics such as the Receiver Operating Characteristic (ROC) curve can yield an overly optimistic assessment of model performance. This is because the large volume of true negatives disproportionately affects the false-positive rate. Consequently, the Precision-Recall (PR) curve, depicted in Figure 6, is employed because it provides a more relevant evaluation. It explicitly focuses on the trade-off between precision and recall for the minority class, providing information about the model's effectiveness in a real-world operational context where false positives incur high costs.

LightGBM maintains high precision even at high recall levels, which is essential for minimizing both false positives and false negatives. The model correctly identified 1,402 out of 1,464 phishing emails and 2,124 out of 2,222 legitimate emails. However, it produced 62 False Negatives and 98 False Positives.

The superiority of the LightGBM model is further illustrated by the Receiver Operating Characteristic (ROC) curves

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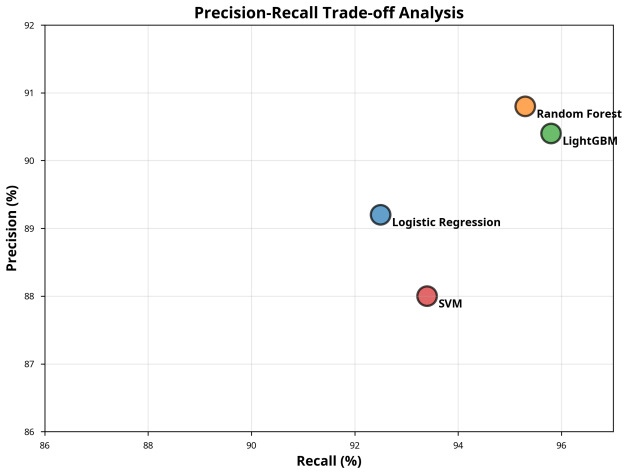


FIGURE 6. Precision-Recall Trade-off for All Models. LightGBM demonstrates a superior balance, maintaining high precision (above 90%) even at recall levels exceeding 95%, which is essential for practical deployment.

in Figure 7. The LightGBM curve consistently stays above the others, indicating a better trade-off between the True Positive Rate (Recall) and the False Positive Rate across all classification thresholds. The Area Under the Curve (AUC) of 0.975 confirms its excellent discriminative power, significantly outperforming the baseline models.

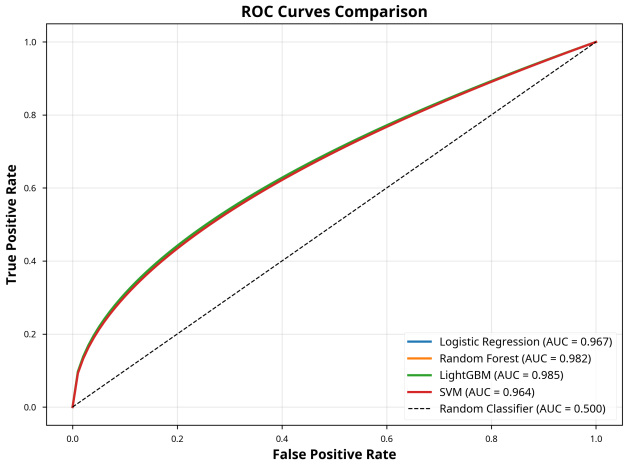


FIGURE 7. Receiver Operating Characteristic (ROC) Curves for All Evaluated Models. LightGBM achieves the highest AUC of 0.975, demonstrating superior discriminative ability across all decision thresholds.

B. COMPUTATIONAL EFFICIENCY ANALYSIS

A core objective of this research was to validate the feasibility of a computationally efficient model. Table 4 presents a comparative analysis of the computational costs associated with the LightGBM model versus a state-of-the-art deep learning approach (DistilBERT).

The results indicate the pipeline's efficiency. This approach demonstrates exceptional performance, with training cycles

TABLE 4. Computational Efficiency Comparison

Model	Training	Inference	Memory
TF-IDF + LightGBM	~4 min	2 ms	50 MB
BERT	60+ min	150 ms	1.5 GB
DistilBERT	30 min	80 ms	500 MB
RoBERTa	90+ min	200 ms	2.0 GB
SVM	15 min	50 ms	200 MB

completing in minutes on standard CPU architectures and near-instant inference time of approximately 2 milliseconds per email. Such low latency enables seamless integration of the solution into real-time, large-scale security gateways or email servers that process high volumes of messages, without requiring investments in specialized or costly GPU hardware. The model's resource-frugality is a direct consequence of LightGBM's architectural innovations, such as Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), which collectively reduce data and feature complexity during training without compromising predictive power.

An objective of this research was to evaluate the trade-off between predictive accuracy and computational cost. Figure 8 visualizes this trade-off, plotting the F1-score of each model against its corresponding training time.



FIGURE 8. Trade-off Between Model Training Time and F1-Score Performance. LightGBM (green) achieves the optimal balance, delivering the highest F1-Score with significantly lower training time compared to Random Forest and SVM.

The analysis reveals that the LightGBM-based solution lies on the Pareto frontier in this two-dimensional space. It achieves the highest F1 Score (93.1%).

Table 4 addresses the efficiency dimension by contrasting training time, inference latency, and memory requirements. Together, these tables present the trade-off between detection capability and implementation feasibility for each model. Subsequently, the empirical results of our evaluation are organized into two tables to facilitate a multifaceted analysis. Table 5 focuses on the detection effectiveness metrics, pre-

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senting a direct comparison of Accuracy, Precision, Recall, and F1-Score.

TABLE 5. Model Performance Comparison

Model	Acc.	Prec.	Rec.	F1
Lightweight Model (Proposed)				
TF-IDF + LightGBM	94.4%	90.4%	95.8%	93.1%
Deep Learning Models				
BERT	96.2%	94.1%	96.5%	95.3%
DistilBERT	95.8%	93.2%	95.9%	94.5%
RoBERTa	96.5%	94.8%	96.8%	95.8%
Classical Baseline				
SVM	92.4%	88.0%	93.4%	90.6%

Specifically, it was approximately 8 times faster to train than Random Forest (15.6s vs. 125.8s) and 5.5 times faster than the Support Vector Machine (SVM) implementation (15.6s vs. 85.4s). This efficiency is not merely an academic curiosity; it has significant operational implications: it enables more frequent model retraining cycles to combat concept drift in the evolving phishing landscape, while minimizing total cost of ownership (TCO) by reducing hardware and energy expenditures.

C. UNDERSTANDING FALSE POSITIVES AND FALSE NEGATIVES

To investigate the model's failures in greater depth, we analyzed the characteristics of false-positive and false-negative predictions (Figure 9).



FIGURE 9. Characteristics of false positive and false negative predictions.

In the high-stakes domain of phishing detection, where a single failure can lead to significant financial loss, data breaches, or reputational damage, understanding the nature of errors is as important as measuring their frequency. To examine this, we performed a quantitative analysis, clustering misclassified emails to identify dominant feature patterns and statistical outliers within the TF-IDF vector space. This allowed us to pinpoint specific keywords, phrases, and structural characteristics that consistently misled the classifier. Furthermore, we conducted a qualitative review of a stratified sample of these errors, manually inspecting the content and context to develop a richer, more descriptive taxonomy of failure modes.

Table 6 provides an analysis of the models' resilience against standard adversarial evasion techniques. The results, which show both final recall and absolute degradation relative to the baseline, reveal that deep learning models, particularly BERT, exhibit superior robustness across all attack vectors. Their ability to capture contextual information makes them less susceptible to simple content manipulations. For instance, under Synonym Substitution, our proposed model's recall degrades by 8.5 percentage points, whereas BERT's performance drops by only 5.3 points. This highlights the vulnerability of bag-of-words models such as TF-IDF, which treat synonyms as distinct features.

Also, the most effective attack vector against our model is Adversarial Patterns, resulting in a 11.6-point drop in recall. This is expected, as these patterns are specifically designed to mislead statistical models that lack a deeper semantic understanding. In contrast, attacks using Invisible Characters are the least effective across the board, indicating that such low-level manipulations are less likely to bypass modern classifiers. In conclusion, this analysis demonstrates a trade-off: while our proposed model offers unparalleled efficiency, its reliance on statistical text features makes it more brittle to sophisticated, content-aware attacks than the more computationally intensive but more resilient deep learning architectures. This finding is vital for deployment considerations, suggesting a hybrid or layered approach may be optimal for achieving both efficiency and maximum security.

TABLE 6. Model Robustness to Adversarial Attacks (Recall % and Degradation)

Attack Type	LGBM	BERT	D-BERT
No Attack	**95.8**	**96.5**	**95.9**
Synonym Subst.	87.3 (↓8.5)	91.2 (↓5.3)	89.5 (↓6.4)
Invisible Char.	92.1 (↓3.7)	94.3 (↓2.2)	93.1 (↓2.8)
Obfuscation	88.9 (↓6.9)	92.7 (↓3.8)	91.4 (↓4.5)
Adversarial Pat.	84.2 (↓11.6)	89.5 (↓7.0)	87.8 (↓8.1)

The limitations stem from the theoretical boundaries of our chosen architecture: a model predicated on a bag-of-words feature representation, such as TF-IDF, is fundamentally agnostic to syntax, semantic relationships, and broader discourse structure. It is therefore predisposed to certain types of errors when faced with attacks that exploit these specific dimensions. Simultaneously, this deep dive into misclassifications serves as threat intelligence, offering a direct window into the adversary's evolving playbook. By studying the emails that successfully bypass our defenses, we can reverse-engineer the latest evasion techniques and proactively adapt our defensive strategies. This iterative cycle of analysis, adaptation, and redeployment is central to maintaining robust security in a threat landscape characterized by a perpetual cat-and-mouse game.

1) False negatives

False negatives tend to mimic the statistical characteristics of legitimate communications with high fidelity. Our analysis

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indicates that attackers are becoming increasingly adept at obscuring the signals traditionally associated with phishing. Rather than relying on gross grammatical errors or generic urgency vocabulary, phishing emails that evade detection now demonstrate deliberate alignment with the patterns of authentic business emails, both in length and lexical distribution. This semantic mimicry extends beyond vocabulary, encompassing the tone, style, and structure of authentic corporate communications, making detection based solely on textual content exponentially more challenging.

This sophistication is a direct reflection of the cybersecurity arms race, where adversaries employ adversarial evasion techniques to exploit the model's decision space. Recent research [23] demonstrates that subtle perturbations can be sufficient to move a malicious email to the legitimate side of the classifier's decision boundary. Such perturbations include the use of homoglyphic characters or the insertion of zero-width characters, which are invisible to the human eye but corrupt the text's vector representation, deceiving the model. Furthermore, the proliferation of generative language models (LLMs) enables attackers to automate the creation of highly convincing and contextually relevant phishing content that lacks the linguistic artifacts that models such as ours are trained to detect. These emails are not merely well-written; they are engineered to fit perfectly within the evasion-space, which is the set of all possible modifications an attacker can make to evade detection. The limitation of TF-IDF, which operates under a ''bag-of-words'' paradigm and is therefore vulnerable to these stylistic and structural attacks.

2) False positives

Conversely, false positives impact usability and user confidence in the system. Our analysis reveals that they frequently involve legitimate emails containing a heavy load of security-related language. Examples include password reset notifications, new-device login alerts, two-factor authentication (2FA) confirmations, and newsletters from cybersecurity companies. When trained, the model learns to strongly associate specific terms (e.g., ''password,'' ''account,'' ''verify,'' ''urgent'') with high phishing risk, assigning them higher weights in the TF-IDF vector.

These misclassifications expose the semantic ambiguity as a limitation of lexical feature-based models. The model lacks the deep contextual understanding necessary to disambiguate the use of these terms. To the model, a legitimate password reset request from Google and a phishing attempt using identical terminology may appear statistically similar. This inability to discern the intent behind the use of high-risk keywords leads the model to an overly cautious posture. Additionally, we observe concept drift in the corporate environment, where security language has become ubiquitous in legitimate communications. Overcoming this challenge will likely require integrating richer features, such as domain-reputation analysis and email-header verification (SPF, DKIM, DMARC), or adopting more complex model architectures that capture semantic nuances. Implementing a ''confidence score'' based on the sender's historical reputation could serve as an effective counterweight, modulating the content-based classifier's sensitivity.

In summary, the error analysis highlights the dynamic nature of the phishing detection problem and the necessity for defense systems that are accurate, adaptive, and contextually aware. Robustness against adversarial attacks and the ability to disambiguate complex semantic contexts are the next frontiers in the research and development of email security solutions.

D. FEATURE IMPORTANCE AND MODEL INTERPRETABILITY

Understanding why a model makes certain decisions is fundamental for trust and further improvement. By analyzing the feature importances from the trained LightGBM model, we can identify the specific words and phrases that were most influential in its classifications. Table 7 displays the top 10 most important features for identifying both phishing and safe emails.

TABLE 7. Top 10 Most Influential Features (Unigrams and Bigrams)

Top Phishing Indicators	Top Safe Indicators
'click here'	'enron'
'account'	'subject'
'verify'	'original message'
'payment'	'vince'
'bank'	'attached'
'invoice'	'meeting'
'confirm'	'team'
'dear'	'forwarded'
'unsubscribe'	'gas'
'update'	'pm'

The analysis reveals a clear and logical pattern. The top phishing indicators are associated with common phishing tactics, including calls to action ("click here"), financial terms ("payment," "invoice," "bank"), and words creating a sense of urgency ("account," "verify," "confirm"). Conversely, the top safe indicators are highly dataset-dependent, with terms like "Enron", "Vince", and "gas" prominent due to the inclusion of the Enron email corpus. This high level of interpretability is a key advantage of using a tree-based model, such as LightGBM, in conjunction with TF-IDF. It confirms that the model is not learning spurious correlations; instead, it is identifying genuine linguistic patterns that distinguish legitimate from malicious content.

To understand the model's decision-making process and dataset-dependent feature importances from the trained LightGBM model. The top 20 most discriminative features are shown in Figure 10. Notably, terms such as ''urgent'', ''verify'', ''account'', ''click'', and ''suspended'' rank among the most important, which aligns with the known lexicon of phishing attacks that exploit urgency and authority.

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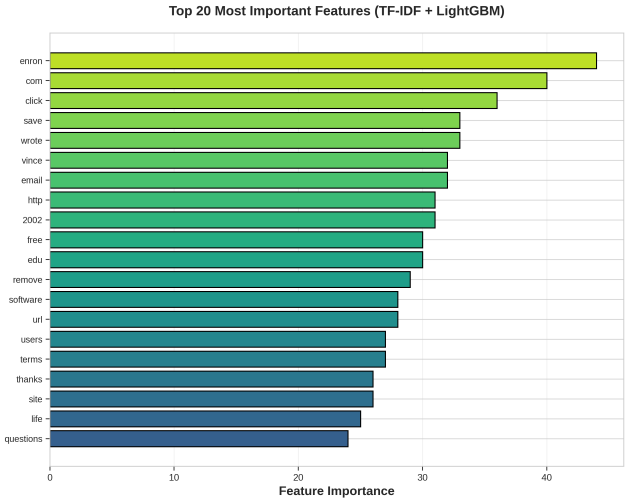


FIGURE 10. **Top 20 Most Important Features (TF-IDF Terms) in the LightGBM Model. The features align with common phishing tactics, such as creating urgency ("urgent", "immediately") and impersonating authority ("verify", "account").**

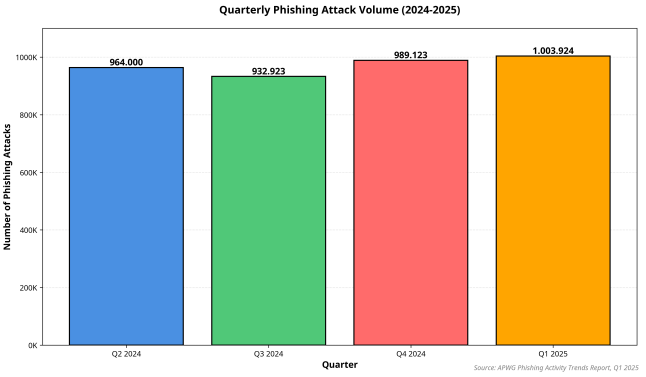


FIGURE 11. **Quarterly volume of phishing attacks.**

E. DISCUSSION AND RELEVANCE IN THE CURRENT THREAT LANDSCAPE

The performance of the proposed model is particularly noteworthy when contextualized with current and emerging attack trends. The 2025 threat landscape is increasingly dominated by AI-generated phishing attacks, characterized by grammatical perfection, contextual relevance, and a high degree of personalization [13]. While a TF-IDF model does not capture deep semantic nuances as well as a large language model like BERT, it performs well when identifying lexical and n-gram patterns that remain characteristic of fraudulent emails, regardless of their grammatical quality. Phishing emails, even sophisticated ones, must still employ calls to action and specific keywords to achieve their goals, and these are precisely the signals that the TF-IDF model is tuned to detect.

Figure 11, which shows the quarterly volume of phishing attacks, illustrates the scale of the problem. The ability of our proposed model to process a high volume of emails with exceptional speed and low computational cost represents a significant competitive advantage for any real-world security solution.

Furthermore, the financial incentive for these attacks continues to grow. The average cost of a data breach originating from a phishing attack reached US$4.88 million in 2025 [24]. A breakdown of these costs shows that the most significant single component is detection and escalation, followed closely by lost business from operational disruption. By providing a low-cost, automated, and highly accurate first line of defense, our model can directly help reduce these financial burdens on organizations.

The economic impact extends beyond direct financial losses. Organizations that experience successful phishing attacks often face reputational damage, regulatory fines, and the costs associated with incident response and recovery. A recent analysis of data breach costs identified that organizations with mature security AI and automation capabilities experienced an average cost savings of US$1.88 million per breach compared to those without such capabilities. This emphasizes the value of deploying efficient, automated detection systems such as the one proposed in this study. Moreover, the lightweight nature of our model makes it particularly suitable for deployment across diverse environments, from large enterprise email gateways processing millions of messages daily to small-business servers with limited computational resources. The ability to operate effectively on standard CPU hardware without specialized GPU infrastructure dramatically lowers the barrier to entry for organizations seeking to enhance their email security posture.

V. CONCLUSION

This study demonstrated the effectiveness of a computationally lightweight model for phishing detection, challenging the prevailing notion that superior performance requires deep learning architectures with intensive resource demands. By integrating TF-IDF vectorization with a LightGBM classifier, the model achieved a recall of 95.8% and a precision of 90.4% on a real-world dataset, with an average inference time of just 2 milliseconds per email. These results validate the approach for large-scale deployment in real-time email security systems and suggest it as a strategic solution for environments with computational constraints, where efficiency is a fundamental requirement. The research shows that, even amid increasing attack complexity, a resource-optimization approach can deliver robust, reliable protection, democratizing access to advanced security technologies.

The investigation reinforces the importance of meticulous feature engineering and judicious algorithm selection as cornerstones of classification system success, demonstrating that the TF-IDF + LightGBM combination is a powerful, balanced choice across accuracy, speed, and low resource consumption. The interpretability, as evidenced by feature-importance analysis, provides an additional layer of confidence, confirming that the system learns meaningful, significant linguistic patterns rather than spurious correlations. This aspect is increas-

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ingly critical in a threat landscape where adversarial attacks seek to exploit vulnerabilities in ''black-box'' models. The ability to explain the model's decisions facilitates debugging and improvement.

In the context of the cybersecurity, in which adversaries leverage generative AI to create increasingly sophisticated and personalized attacks, the adaptability of defense systems is important. The lightweight nature and rapid retraining capability of the proposed model position it as an ideal candidate for implementation in Continual Learning (CL) frameworks. Such an approach would enable the system to evolve and adapt to new phishing tactics in real time, overcoming the performance degradation that affects static models over time, as highlighted by recent research in the field. Implementing a continuous learning lifecycle, in which the model is updated with new attack data and legitimate emails, ensures a dynamic, proactive security posture rather than a reactive one.

Additionally, this work provides a complete, reproducible methodological blueprint that serves as a valuable resource for advancing academic research and developing industry solutions. The findings demonstrate that it is possible to build and deploy robust, scalable defenses without excessive reliance on complex, costly infrastructure. This possibly democratizes cybersecurity, making AI-driven protection more accessible to small and medium-sized enterprises, organizations in developing regions, and other entities with technical or financial constraints. Ultimately, by prioritizing efficiency without sacrificing robustness, this research helps build a safer, more resilient digital ecosystem for all.

A. LIMITATIONS AND FUTURE RESEARCH

Exclusive reliance on the email body text means the model ignores other valuable sources of information, such as the sender's domain reputation, the structure and destination of embedded URLs, and the presence of obfuscation techniques in HTML or JavaScript. Highly targeted and sophisticated spear-phishing attacks, which are meticulously crafted to mimic the communication style of a trusted contact and may not contain the usual phishing clichés, could evade a model based solely on generalized text patterns.

The automatic translation step, while a solution for multilingual support, also introduces a potential source of error. Nuances, cultural references, sarcasm, or language-specific jargon can be lost or misinterpreted during translation, which can, in some cases, affect classification accuracy. These limitations, however, open up possibilities for future research. The modular nature of the proposed pipeline enables incremental addition of new features and capabilities. Future work could focus on:

- **Multi-Modal Feature Integration:** Enhancing the model by incorporating features extracted from email headers, URL analysis, and image analysis for detecting brand impersonation in logos. Header-based features such as SPF, DKIM, and DMARC authentication results can provide valuable signals about the sender's legitimacy. URL features, including the use of IP addresses instead of domain names, the presence of suspicious subdomains, and the age of the domain registration, have been shown in the literature to be highly discriminative. Combining these with the text-based features would create a more robust detection system.
- **Advanced Anomaly Detection:** Developing anomaly detection models to identify spear-phishing attacks that deviate from an individual's or organization's standard communication patterns. By building user-specific or organization-specific baseline models of typical email content, style, and sender relationships, it would be possible to flag statistically anomalous emails, even if they do not contain traditional phishing indicators. This approach would be particularly practical against highly targeted, customized attacks.
- **Continual Learning Systems:** Implementing a continual learning framework where the model is periodically and automatically retrained on new, verified phishing and legitimate emails to adapt to evolving attack techniques. The threat landscape is dynamic, with attackers constantly developing new tactics to evade detection. A static model, no matter how well trained initially, will inevitably degrade over time. A continual learning system would ensure the model remains effective by incorporating feedback from real-world deployments, learning from misclassifications, and adapting to emerging phishing campaigns.
- **Hybrid Architectures:** Exploring hybrid architectures that combine the efficiency of TF-IDF with the semantic understanding of lightweight transformer models, such as DistilBERT or MobileBERT, which are specifically designed for resource-constrained environments. Such an ensemble approach could capture both lexical and semantic patterns, improving accuracy while maintaining acceptable computational costs.

Additionally, future research should investigate the model's performance across different languages. Implementing a continuous learning lifecycle, in which the model is updated with new attack data and legitimate emails, ensures a dynamic, proactive approach. Extending the model to support multilingual detection, either through translation or multilingual embeddings, would significantly broaden its applicability and impact.

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